SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Dharshan B (192312150) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Dharshan.

level -3

CSA1714 - Aritificial Intelligence

CO1 - AT1 - Analytical Problem Solving

1) vaccum cleaner Robot - PEAS Description and Task Environment

PEAS Description

Component	Description
performance measure	clean both rooms, minimize time, minimize energy usage, avoid unnecessary movements
Environment	Two rooms, each can be either clean or Dirty.
Actutators	Move left, Move Right, suck (clean), No operation (stop).
Sensors	Dirt Sensor, location Sensor

Task Environment:

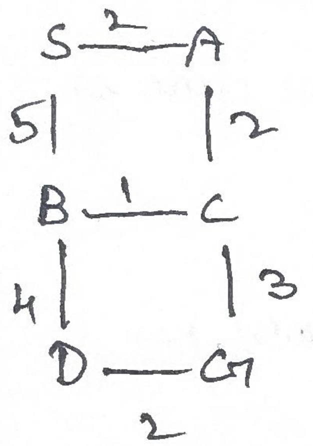
Characteristic	Description
observability	Fully observable - The robot knowns it current location and whether the room is clean
Determinism	Each action has a predictable outcome
Episodicity	sequential - current actions affect future States.
Number of Agents	single Agent - only one vaccum robot performs

Conclusion:

The vackum-cleaner agent is single-agent, fully observable, deterministic and sequential intelligent agent that cleans both rooms efficiently.

Uniform Cost Search (UCS).

Graph:-



EDGE Cost:-

* $S \to A = 2$
* $S \to B = 5$
* $A \to C = 2$
* $B \to C = 1$
* $B \to D = 4$
* $C \to G = 3$
* $D \to G = 2$

Step-by-step UCS

Step	Frontier (cost)	Explored
Start	S(0)	{ }
1	A(2), B(5)	S
2	C(4), B(5)	S, A
3	B(5), G(7)	S, A, C
4	D(9), G(7)	S, A, C, B
5	Goal G(7) Found	S, A, C, B

Least Cost path :-

$S \to A \to C \to G = 2+2+3 \Rightarrow 7$

Conclusion:-

Uniform cost search always expands the node with the lowest cumulative path cost, guaranteeing the optimal solution when all edge costs are non-negative.

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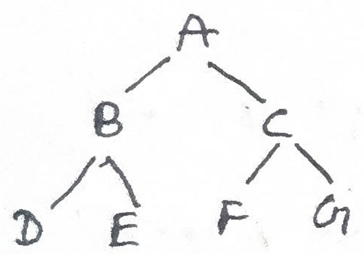
3) Iterative Deepening Search (IDS):-

It is repeatedly performs Depth-First Search (DFS) with increasing depth limits until the goal node is found.

Graph:-



Depth limit = 0

Visit only the root. → A

Visit order : A

Goal Found ? No

Depth limit = 1

Visit order : A → B → C

Goal Found ? No.

Depth Limit = 2

visit order:

A → B → D → E → C → F → G

Goal Found?

Yes

Advantage of IDS :-

* Requires very little memory
* Guarantees finding the shallowest goal
* Suitable for large search spaces.
* Complete for finite search trees.

Conclusion :-

Iterative Deepening Search combines space efficiency of DFS with the completeness and optimal shallow-solution property of BFS, making it an effective search strategy when the goal depth is unknown.